

Chapter 5 homework

Setup

```
library(tidyverse)
library(here)
library(broom)
library(pwr)
library(psych)

knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)
```

General notes on the model-comparison workflow

For most questions below, you will compare:

- **Model C (constrained / null model)**: a model where the intercept is fixed (often to 0 or to a known/reference value).
- **Model A (augmented model)**: a model where the intercept is estimated from the data.

For a one-parameter intercept-only Model A,

$$\text{Model A: } Y_i = b_0 + \varepsilon_i$$

and for a fixed-intercept Model C (often b_0 fixed to some value c),

$$\text{Model C: } Y_i = c + \varepsilon_i.$$

The key quantities:

- SSE is sum of squared errors.

- $\text{PRE} = \frac{\text{SSE}_C - \text{SSE}_A}{\text{SSE}_C}$.
- For a one-parameter comparison (typically $P_A - P_C = 1$), the usual model-comparison F statistic is:

$$F = \frac{(\text{SSE}_C - \text{SSE}_A)/(P_A - P_C)}{\text{SSE}_A/(n - P_A)}.$$

Part A: Hand-calculation practice (small, self-contained datasets)

A1. Schizophrenia age of onset

Assume the mean age of onset of schizophrenia in a sample of $n = 50$ persons was $\bar{y} = 21.5$ years and the variance was $s^2 = 2.5$ (years²).

1. Specify the model that uses the mean to describe the data (Model A), in both words and in equation form.

$$Y_i = 21.5 + \varepsilon_i$$

2. What is the SSE for this model?

(Use the definitional relationships between variance, sums of squares, and SSE for an intercept-only model.)

```
sse <- ((50 - 1) * 2.5)
sse
```

```
[1] 122.5
```

A2. Multiculturalism vs. colorblindness (paired data)

Ryan, Hunt, Weible, Peterson, and Casas (2007) investigated Americans' interethnic ideologies. They asked college students to indicate the extent to which they believed that multiculturalism and colorblind ideologies would improve interethnic relations in the U.S.

Assume you have data for 10 Black participants:

Subjid	MC	CB	Diff (MC – CB)
1	5.0	4.0	1.0
2	4.0	3.5	0.5
3	6.0	4.0	2.0
4	6.0	6.0	0.0
5	5.0	4.5	0.5
6	4.5	5.0	-0.5
7	5.0	3.0	2.0
8	6.5	4.5	2.0
9	7.0	5.0	2.0
10	6.5	3.0	3.5

A2a. By hand

1. Compute **by hand** the mean, variance, and standard deviation of the difference variable. Use definitional formulas.

Mean = 1.3 Variance = 1.456 Standard deviation = 1.21

2. Compute SSE_C for the simple model that predicts no difference (i.e., that the difference is 0). This is Model C.

$SSE_C = 30$

3. Compute SSE_A for the simple model that best describes the differences based on the data (Model A).

$SSE_A = 13.1$

4. Compute PRE and F .

PRE = 0.563 $F = 11.611$

5. Specify the null hypothesis tested by comparing these two models.

$$\Delta_i = 0 + \varepsilon_i$$

6. Indicate whether you would reject or fail to reject Model C / the null hypothesis.

PRE* = 0.362 $F^* = 5.117$

7. Write up your findings in journal style.

In this sample from Ryan, Hunt, Weible, Peterson, and Casas' (2007) study, the average Black college student believed that multiculturalism would better improve interethnic relationships in the US than colorblind ideologies, with an average difference of 1.3. Given the observed F statistic of 11.611 and the critical F statistic of 5.117, we reject the null hypothesis that there is no difference in attitudes on multiculturalism and colorblind ideologies.

(You can optionally use R at the end to check arithmetic, but do the main work by hand.)

A2b. In R (reproducing your values)

Recreate the analysis in R and verify the quantities above (mean, variance/SD, SSE_C , SSE_A , PRE, F , and p).

```
subjid <- c(1:10)
mc <- c(5, 4, 6, 6, 5, 4.5, 5, 6.5, 7, 6.5)
cb <- c(4, 3.5, 4, 6, 4.5, 5, 3, 4.5, 5, 3)

df <- data.frame(subjid, mc, cb)

df <- df |>
  mutate(diff = mc - cb)

m <- df |>
  summarize(mean = mean(diff))
m
```

```
mean
1  1.3
```

```
variance <- df |>
  summarize(variance = var(diff))

sd <- sqrt(variance)

df <- df |>
  mutate(diff_dev = mc - cb)

model_c <- lm(diff_dev ~ 0, data = df)
model_a <- lm(diff_dev ~ 1, data = df)

sse_c <- deviance(model_c)
sse_a <- deviance(model_a)
```

```
pre <- ((sse_c - sse_a) / sse_c)

pa <- 1
pc <- 0

n <- 10

fstat <- (pre / (pa - pc)) / ((1 - pre) / (n - 1))

p_val <- 1 - pf(fstat, 1, 9)

m
```

```
mean
1 1.3
```

```
variance
```

```
variance
1 1.455556
```

```
sd
```

```
variance
1 1.206464
```

```
pre
```

```
[1] 0.5633333
```

```
fstat
```

```
[1] 11.61069
```

```
p_val
```

```
[1] 0.00778102
```

A2c. Confidence interval

Construct a 95% confidence interval for the multiculturalism – colorblindness difference score in this sample.

```
summary(model_a)
```

Call:

```
lm(formula = diff_dev ~ 1, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8	-0.8	0.2	0.7	2.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.3000	0.3815	3.407	0.00778 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.206 on 9 degrees of freedom

```
1.3 + (2 * .3815)
```

```
[1] 2.063
```

```
1.3 - (2 * .3815)
```

```
[1] 0.537
```

```
0.537 <-> 2.063
```

A2d. Power planning with pwr

A researcher wants to replicate the finding with a new sample of 10 participants.

1. Use results from this sample and the **pwr** package to estimate the number of participants needed to replicate your observed effect (at $\alpha = .05$). $n = 47$

2. Use the same package to estimate the number of participants needed to detect a **medium** effect size (according to Cohen's very arbitrary definition...). $n = 24$

```
groups <- 2
obs <- 10
within_var <- var(df$diff) # estimate
between_var <- var(c(0, 1)) * ((groups - 1) / groups) # population

power.anova.test(groups = groups,
                  between.var = between_var,
                  within.var = within_var,
                  sig.level = .05,
                  power = .8)
```

Balanced one-way analysis of variance power calculation

```
groups = 2
n = 46.67788
between.var = 0.25
within.var = 1.455556
sig.level = 0.05
power = 0.8
```

NOTE: n is number in each group

```
power.anova.test(groups = groups,
                  between.var = between_var,
                  within.var = within_var,
                  sig.level = .05,
                  power = .5)
```

Balanced one-way analysis of variance power calculation

```
groups = 2
n = 23.35765
between.var = 0.25
within.var = 1.455556
sig.level = 0.05
power = 0.5
```

NOTE: n is number in each group

A3. Gym BMI (one-sample mean test)

A researcher wonders whether people who go to the gym in January have above-normal BMIs, that is, above 24.9 (upper level of “normal,” according to the CDC). She obtains the BMI scores below for a randomly selected group of 10 gym users:

Participant	BMI
1	24.0
2	27.0
3	25.0
4	27.0
5	27.0
6	23.0
7	22.0
8	30.0
9	26.0
10	29.0

1. Specify Models C and A and the null hypothesis for this statistical test. What is this test commonly called?

$$\text{Model A: } Y_i = 26 + \varepsilon_i$$

$$\text{Model C: } Y_i = 24.9 + \varepsilon_i.$$

$$\text{Null Hypothesis } (H_0) : b_0 = 24.9$$

2. Conduct the analysis **by hand**. mean = 26 predicted mean = 24.9 sse for model c = 70.1 sse for model a = 58 pre = 0.173
3. Conduct the analysis in R.
4. Provide a statistical conclusion.
5. Write up your findings in journal style.

```
id <- c(1:10)
bmi <- c(24.0, 27.0, 25.0, 27.0, 27.0, 23.0, 22.0, 30.0, 26.0, 29.0)

d <- data.frame(id, bmi)
```



```
d <- d |>
  mutate(bmi_dev = bmi - 24.9)

m_c <- lm(bmi_dev ~ 0, data = d)
m_a <- lm(bmi_dev ~ 1, data = d)

summary(m_c)
```

Call:

```
lm(formula = bmi_dev ~ 0, data = d)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.90	-0.65	1.60	2.10	5.10

No Coefficients

Residual standard error: 2.648 on 10 degrees of freedom

```
summary(m_a)
```

Call:

```
lm(formula = bmi_dev ~ 1, data = d)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.00	-1.75	0.50	1.00	4.00

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.1000	0.8028	1.37	0.204

Residual standard error: 2.539 on 9 degrees of freedom

```
ssec <- deviance(m_c)
ssea <- deviance(m_a)

pre_2 <- ((ssec - ssea) / ssec)
```

```

pa <- 1
pc <- 0

n_2 <- 10

fstat_2 <- (pre_2 / (pa - pc)) / ((1 - pre_2) / (n_2 - 1))

pval <- 1 - pf(fstat_2, 1, 9)

pre_2

```

```
[1] 0.1726106
```

```
fstat_2
```

```
[1] 1.877586
```

```
pval
```

```
[1] 0.2038144
```

```

qpre <- function(p,
                 df1,
                 df2) {

  f <- qf(p = p,
         df1 = df1,
         df2 = df2)

  crit_pre <- f / (f + (df2/df1))

  return(crit_pre)

}

pre_crit <- qpre(p = .95, 1, 9)
f_crit <- qf(1 - .05, 1, 9)

pre_crit

```

```
[1] 0.3624868
```

```
f_crit
```

```
[1] 5.117355
```

Given the critical F and pre values of 5.117 and 0.382 respectively, we fail to reject the null hypothesis that people who go to the gym in January have normal BMIs.

Part B: Fitness dataset (model comparison, confidence intervals, power)

B0. Read the data

Read in the `fitness.csv` dataset from the course project `data/` folder.

```
fitness <- read_csv(here("data", "fitness.csv"))
```

B1. Resting pulse vs. “normal” (72 bpm)

The variable `RSTPULSE` contains resting pulse rate for each participant in the fitness study. The normal pulse rate for men is 72.

B1a. 95% CI for b_0 in Model A

Construct a 95% confidence interval for b_0 in Model A.

- Do this **by hand** using a critical value (e.g., $t_{\alpha/2, df}$), and then confirm with R.
- Make note of what in R output corresponds to your hand calculation.

For reference, the by-hand CI for an intercept estimate is:

$$\hat{b}_0 \pm t_{\alpha/2, df} \cdot SE(\hat{b}_0).$$

```
sd <- 8.294
se <- ((sd)/sqrt(31))

upper <- 53.7 + (2 * se)
lower <- 53.7 - (2 * se)

upper
```

```
[1] 56.67929
```

```
lower
```

```
[1] 50.72071
```

```
50.721 <-> 56.679
```

B1b. Journal-style summary

Write a complete summary:

1. briefly explain the data;
2. set up the analysis (describe the comparison without using “Model A/C” language if you prefer);
3. report mean (and SD), test statistic (t or F), PRE, p , and the confidence interval; and
4. state statistical and substantive conclusions.

```
fitness <- fitness |>
  mutate(rstpulse_dev = RSTPULSE - 72)

modelc <- lm(rstpulse_dev ~ 0, data = fitness)
modela <- lm(rstpulse_dev ~ 1, data = fitness)

summary(modelc)
```

Call:

```
lm(formula = rstpulse_dev ~ 0, data = fitness)
```

Residuals:

```
    Min      1Q  Median      3Q     Max
```

```
-32.0 -24.0 -20.0 -13.5 4.0
```

No Coefficients

Residual standard error: 20 on 31 degrees of freedom

```
summary(modela)
```

Call:

```
lm(formula = rstpulse_dev ~ 1, data = fitness)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.742	-5.742	-1.742	4.758	22.258

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-18.26	1.49	-12.26	3.29e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.294 on 30 degrees of freedom

```
fitness |>  
  summarize(mean = mean(RSTPULSE))
```

```
# A tibble: 1 x 1
```

```
  mean
```

```
<dbl>
```

```
1  53.7
```

```
c_sse <- deviance(modelc)
```

```
a_sse <- deviance(modela)
```

```
c_sse
```

```
[1] 12398
```

```
a_sse
```

```
[1] 2063.935
```

```
fit_pre <- (c_sse - a_sse) / c_sse  
fit_pre
```

```
[1] 0.8335267
```

```
fit_true <- fit_pre / ((1) / ((1 - fit_pre) / (30)))
```

B1c. Estimate of true PRE

If you wanted to replicate this study, what would be your best guess for the *true* value of PRE?

```
true_pre <- 1 - ((1 - fit_pre) * ((1/30)))  
true_pre
```

```
[1] 0.9944509
```

B1d. Power for a replication at the same n

Given your estimate of true PRE, if you replicated with the same sample size, what would be your power to detect the effect? Use `pwr.f2.test()`.

A common translation between PRE and Cohen's f^2 for this model-comparison setup is:

$$f^2 = \frac{\text{PRE}}{1 - \text{PRE}}.$$

```
f_sqr <- (fit_pre / (1 - fit_pre))  
f_sqr
```

```
[1] 5.006971
```

```
pwr.f2.test(1, 30, 5, sig.level = .05)
```

Multiple regression power calculation

```
u = 1
v = 30
f2 = 5
sig.level = 0.05
power = 1
```

Power = 1

B2. Did running increase pulse rate? (within-person change)

Using the same dataset, compare resting pulse (RSTPULSE) to post-run pulse (RUNPULSE). Did running increase pulse rate?

Hint: create a new variable such as:

$$\text{change} = \text{RUNPULSE} - \text{RSTPULSE}.$$

B2a. 95% CI for b_0 in Model A

Construct a 95% CI for b_0 in your Model A for the change score. Do it by hand, then confirm with R. mean = 116 variance = 120 sd = 10.9 se = 1.96

112.08 <-> 119.94

```
fitness <- fitness |>
  mutate(pulse_change = RUNPULSE - RSTPULSE)

fitness |>
  summarize(mean = mean(pulse_change),
            variance = var(pulse_change),
            sd = sqrt(variance))
```

```
# A tibble: 1 x 3
  mean variance    sd
  <dbl>     <dbl> <dbl>
1  116.       120.  10.9
```

B2b. Estimate of true PRE

If you wanted to replicate this study, what would be your best guess for the true value of PRE?

```
model_0 <- lm(pulse_change ~ 0, data = fitness)
model_1 <- lm(pulse_change ~ 1, data = fitness)

sse0 <- deviance(model_0)
sse1 <- deviance(model_1)

pulse_pre <- (sse0 - sse1) / sse0
true_pulse_pre <- 1 - ((1 - pulse_pre) * ((1/30)))
true_pulse_pre
```

```
[1] 0.9997147
```

B2c. Required effect size for power = .80 at $n = 100$

How big would true PRE need to be to yield power = .80 in a future study with $n = 100$ participants? Use `pwr.f2.test()`.

```
pwr.f2.test(1, 99, sig.level = 0.05, power = 0.8)
```

Multiple regression power calculation

```
u = 1
v = 99
f2 = 0.07925829
sig.level = 0.05
power = 0.8
```



```
f_2 <- 0.079

pre_pulse <- .079/1.079
pre_pulse
```

```
[1] 0.07321594
```

B3. Oral contraceptives and systolic blood pressure (Rosner example)

Ten women are studied to determine if oral contraceptives increase systolic blood pressure (mm Hg). Baseline blood pressure is measured before and then again several months after oral contraceptives are started. The following **changes** in systolic blood pressure are observed:

13, 3, -1, 9, 7, 7, 6, 4, -2, 2

A summary table (from SAS) reports:

Mean	Std Dev	USS	CSS	T	Prob >
4.8000000	4.5655716	418.0000000	187.6000000	3.3246511	0.0089

Here, **USS** (“uncorrected sum of squares”) represents squared error around the model $Y_i = 0 + \varepsilon_i$ (i.e., Model C with intercept fixed at 0). **CSS** (“corrected sums of squares”) represents squared error around $Y_i = b_0 + \varepsilon_i$ (Model A with estimated intercept).

B3a. Interpret the output using model comparison

Use this output to determine whether blood pressure significantly increased after oral contraceptives were begun.

1. Specify Models A & C (equations are encouraged).
2. Specify the null hypothesis.
3. Calculate PRE and F .
4. State your statistical and substantive one-sentence conclusions.

Model C:

$$\text{RSTPULSE}_i = 0 + \varepsilon_i$$

Model A:

$$\text{RSTPULSE}_i = 4.8 + \varepsilon_i$$

Null hypothesis:

$$H_0 : b_0 = 0$$

```
cont_pre <- (418 - 187.6) / 418
cont_pre
```

```
[1] 0.5511962
```

```
f_cont <- (cont_pre / (1)) / ((1 - cont_pre) / (9))
f_cont
```

```
[1] 11.0533
```

```
critical_f <- qf(.95, 1, 9)
critical_f
```

```
[1] 5.117355
```

Because the F statistic is greater than the critical F value of 5.11, we can reject the null hypothesis.

B3b. Estimate of true PRE

If you wanted to replicate this study, what would be your best guess for the true value of PRE?

```
true_cont_pre <- 1 - ((1 - cont_pre) * ((1/9)))
true_cont_pre
```

```
[1] 0.9501329
```

B3c. Sample size for power targets

What sample size would you need to achieve power = .80? What sample size would you need to achieve power = .90? Use `pwr.f2.test()`.

```
pwr.f2.test(1, f2 = .901, sig.level = .05, power = .8)
```

Multiple regression power calculation

```
u = 1
v = 9.007769
f2 = 0.901
sig.level = 0.05
power = 0.8
```

```
pwr.f2.test(1, f2 = .901, sig.level = .05, power = .9)
```

Multiple regression power calculation

```
u = 1
v = 11.89708
f2 = 0.901
sig.level = 0.05
power = 0.9
```

n = 10 n = 13

B4. Power and design choices (GPA example)

In a particular year, a graduate program admitted 24 new students. The admissions committee wants to interpret the students' first-year GPA.

B4a. Approximate power for a “moderate” effect

Graduate students in this program are required to maintain a 3.0 average. The committee plans to compare:

- Model C: GPA fixed at 3.0
- Model A: GPA estimated as the students' mean

If the true difference corresponds to what the text refers to as a “moderate” effect in the social sciences, what are the chances that the committee will correctly conclude Model A is better than Model C if they use $\alpha = .05$?

1. Calculate approximate power using the book approach.
2. Verify using `pwr` (it will not match exactly, but should be close).

```
pwr.anova.test(k = 2, n = 24, f = .5, sig.level = .05)
```

Balanced one-way analysis of variance power calculation

```

k = 2
n = 24
f = 0.5
sig.level = 0.05
power = 0.923778

```

NOTE: n is number in each group

B4b. Two ways to increase power without increasing n

Given that they cannot admit additional students, state two things they could do to increase power.

An alternative way to increase power would be to increase observations of students' GPAs or to decrease the level of statistical significance at which they seek to estimate.

B4c. Try-out question: setting up a conditional model comparison

The committee wants to know whether GRE-Verbal, GRE-Math, and undergraduate GPA are useful predictors of first-year GPA.

We have not yet learned how to *test* these conditional models, but set up the comparison:

- Model C: predicts the mean
- Model A: predicts GPA conditionally using all 3 predictors

Specify both models and indicate $n - P_A$ and $P_A - P_C$ for this comparison.

Model A: $GPA_i = b_0 + a(GREVerbal_i) + b(GREMath_i) + c(UndergradGPA) + \varepsilon_i$

Model C: $GPA_i = 3.0 + \varepsilon_i$.

Null Hypothesis (H_0) : $b_0 = 3.0$

$n - P_A = 20$ $P_A - P_C = 3$

Part C: Your own data (leave code blocks empty)

These questions refer to the model comparison you tested previously using your own dataset.

C1. Confidence interval for b_0

Identify the 95% confidence interval for b_0 in Model A. Explain what this interval represents.

```
gss2024 <- readRDS(file = here::here("data", "gss2024.rds"))
```

```
gss <- gss2024 |>
  select(courtsnv) |>
  drop_na() |>
  haven::zap_labels()
```

```
gss_recoded <- gss |>
  mutate(
    courtsdv = recode(courtsnv,
                      `1` = -1,
                      `2` = 1,
                      `3` = 0)
  )
```

```
gss_recoded |>
  summarize(
    n = n(),
    mean_courts = mean(courtsdv, na.rm = TRUE),
    sd_courts = sd(courtsdv, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 3
      n mean_courts sd_courts
<int>   <dbl>   <dbl>
1   616     0.542     0.841
```

```
gss_recoded <- gss_recoded |>
  mutate(dev = courtsdv - 0)

mod_c <- lm(dev ~ 0,
            data = gss_recoded)
```

```
mod_a <- lm(dev ~ 1,
            data = gss_recoded)

summary(mod_c)
```

Call:

```
lm(formula = dev ~ 0, data = gss_recoded)
```

Residuals:

Min	1Q	Median	3Q	Max
-1	1	1	1	1

No Coefficients

Residual standard error: 1 on 616 degrees of freedom

```
summary(mod_a)
```

Call:

```
lm(formula = dev ~ 1, data = gss_recoded)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.5422	0.4578	0.4578	0.4578	0.4578

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.54221	0.03388	16	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8409 on 615 degrees of freedom

```
sse_c_own <- deviance(mod_c)
sse_a_own <- deviance(mod_a)
pre_own <- (sse_c_own - sse_a_own) / sse_c_own
pre_own
```

```
[1] 0.2939893
```

```
f_own <- (pre_own / (1)) / ((1 - pre_own) / (615))  
f_own
```

```
[1] 256.0916
```

C2. Journal-style summary

Write a complete journal-style summary for your data.

```
critical_pre_own <- qpre(p = .95, 1, 615)  
f_critical_own <- qf(.95, 1, 615)  
  
critical_pre_own
```

```
[1] 0.006231853
```

```
f_critical_own
```

```
[1] 3.856624
```

Given the critical F statistic and PRE value of 2.85 and 0.006 respectively, we can reject the null hypothesis that opinions are evenly distributed between “too harshly” and “not harshly enough”. ## C3. Estimate of true PRE

If you wanted to replicate this study, what would be your best guess for the true value of PRE?

```
true_pre_own <- 1 - ((1 - pre_own) * ((1/615)))  
true_pre_own
```

```
[1] 0.998852
```

C4. Sample size for power = .80

Given your estimate of true PRE, what sample size would you need to achieve power = .80? Use `pwr.f2.test()`.

```
f2_own <- true_pre_own / (1 - true_pre_own)
# pwr.f2.test(u = 1, f2 = f2_own, power = .0) ## Unsure why this is resulting in an error
```

C5. Sketch the PRE distributions

Draw a chart that shows the distribution of PRE under both the H_0 and H_1 hypotheses. Label α , β , power, and the surprise point.

```
# Unsure what this is asking -- will ask in class
```